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BSIT – 4E

Enterprise Infrastructure

Part 1 — Company Profile

1. Company Name

Tyr Technologies, Inc.

Tyr Technologies, Inc. is a newly established software development and IT solutions startup that specializes in creating modern digital solutions for businesses and organizations.

2. Nature of Business

Tyr Technologies Inc. is a technology company specializing in software development, business systems, web applications, database solutions, and IT services.

The company develops customized digital solutions designed to help organizations automate business processes, manage information, improve productivity, and provide better services to their customers.

Primary Services

- Custom software development
- Web and mobile application development
- Business management systems
- Database design and management
- Cloud-based solutions
- Business process automation
- IT infrastructure support
- Software maintenance and technical support

As a startup, Tyr Technologies will initially operate from a single office floor with 20 employees distributed among Information Technology, Human Resources, Finance, and Sales.

3. Company Vision

To become a trusted technology company that transforms businesses through innovative, secure, and reliable digital solutions.

4. Company Mission

Tyr Technologies, Inc. is committed to developing practical and innovative software solutions that help organizations improve their operations, productivity, and customer experience. The company aims to deliver secure, reliable, and user-focused technologies while continuously improving its services to meet the evolving needs of its clients.

5. Office Location

Tyr Technologies, Inc.

2nd Floor, Tyr Technology Center

National Highway, Santa Cruz, Laguna, Philippines

The company occupies a single office floor designed for 20 employees. The office contains dedicated workspaces for each department, a server/network area, meeting spaces, and shared facilities.

The IT infrastructure will be centrally managed to provide employees with reliable internet access, secure internal resources, file storage, communication services, and access to business applications.

6. Organizational Structure

Tyr Technologies, Inc. will use a functional organizational structure. Employees are grouped into departments based on their responsibilities and areas of expertise.

Sales Department

Department Responsibilities

Information Technology Department

- Develops and maintains company software.
- Manages servers, computers, networks, and IT infrastructure.
- Provides technical support to employees.
- Maintains cybersecurity and system availability.

Human Resources Department

- Handles recruitment and employee onboarding.
- Maintains employee records.
- Manage company policies and employee-related processes.

- Supports employee development.

Finance Department

- Handles accounting and financial records.
- Manages payroll and company expenses.
- Prepare financial reports.
- Assists with budgeting and financial planning.

Sales Department

- Handles customer acquisition and sales.
- Maintains client relationships.
- Identify new business opportunities.
- Coordinates with IT regarding customer requirements.

7. Employee Distribution

Tyr Technologies Inc. will initially have 20 employees distributed among four departments. The distribution follows the project scenario: 5 employees in Information Technology, 4 in Human Resources, 5 in Finance, and 6 in Sales.

Department	Employees	Primary Responsibilities
Information Technology	5	Software development, infrastructure, systems administration, technical support
Human Resources	4	Recruitment, employee records, administration
Finance	5	Accounting, payroll, budgeting, financial management
Sales	6	Sales, customer relations, business development
Total	20	Overall company operations

Proposed Positions

Department	Position	Number
Information Technology	IT/System Administrator	1

Department	Position	Number
Information Technology	Software Developers	3
Information Technology	IT Support Technician	1
Human Resources	HR Manager	1
Human Resources	HR Officers	2
Human Resources	HR Assistant	1
Finance	Finance Manager	1
Finance	Accountants	2
Finance	Finance Officers	2
Sales	Sales Manager	1
Sales	Sales Representatives	4
Sales	Business Development Officer	1
Total		20

Company Profile Summary

Tyr Technologies, Inc. is a newly established 20-employed software development and IT solutions startup operating from a single office floor in Santa Rosa, Laguna. The company provides software development, business systems, database solutions, cloud services, and technical support to its clients.

The organization consists of four departments: Information Technology, Human Resources, Finance, and Sales. The IT department is responsible for developing the company's technology solutions while also maintaining the organization's internal IT infrastructure.

Because Tyr Technologies currently has no computers, servers, network infrastructure, internet infrastructure, or security policies, the company requires a complete IT infrastructure plan before operations begin.

This infrastructure plan will provide Tyr Technologies with a reliable foundation for its current operations while allowing the company to expand its workforce, services, and technology infrastructure in the future.

Part 2 — Enterprise Hardware Inventory

Hardware	Quantity	Department	Purpose
Desktop Computers	15	IT, HR, Finance, Sales	Main computers for office work
Laptops	5	IT, Sales	Portable computers for work and meetings
Server	1	IT	Stores files, applications, and company services
Router	1	IT	Connects the office network to the internet
Managed Switch	1	IT	Connects computers and other wired devices
Network Printer	2	HR, Finance	Printing documents and reports
UPS	3	IT	Protects important equipment from power interruptions
Wireless Access Point	2	IT	Provides Wi-Fi throughout the office
NAS Storage	1	IT	Stores shared files and backups
External Backup Drive	2	IT	Keeping extra copies of important data
Monitors	20	All Departments	Provides a display for each employee

I chose this hardware components based on the needs of Tyr Technologies and its 20 employees. The company needs enough computers for employees to perform their daily tasks, while the IT department needs additional equipment for managing the company's systems and network.

The server and NAS will be useful for storing company files and important data. The router, switch, and access points will provide the internet and network connection needed by the employees.

I also included printers for departments that may need to print documents regularly. The UPS units can help prevent problems when there is sudden power interruption, while the external drives can be used to keep backup copies of important files.

Overall, these hardware choices provide Tyr Technologies with the basic equipment needed to start its operations.

Part 3 — Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	Current	Paid	Main operating system for employee computers

Software	Version	License	Purpose
Ubuntu Server	24.04 LTS	Free / Open Source	Used for the company server and IT services
Microsoft Office	Microsoft 365	Subscription	Used for documents, spreadsheets, presentations, and other office tasks
Visual Studio Code	Current	Free	Used by developers for writing and editing code
Git	Current	Free / Open Source	Used for managing and tracking code changes
GitHub Desktop	Current	Free	Make it easier to manage Git repositories
VirtualBox	Current	Free / Open Source	Used for creating and testing virtual machines
Google Chrome	Current	Free	Main web browser for work and online services
Microsoft Defender	Built-in	Included with Windows	Provides basic antivirus and security protection
AnyDesk	Current	Free / Paid	Used for remote technical support
7-Zip	Current	Free / Open Source	Used for compressing and extracting files

These software programs were selected based on the different tasks that employees at Tyr Technologies will perform. **Windows 11 Pro and Microsoft 365** will be used for normal office work, while **VS Code, Git, and GitHub Desktop** will support the developers.

Ubuntu Server will be used for the company's server, and **VirtualBox** can be used by the IT department for testing different operating systems and configurations. **Chrome** will be used for accessing websites and online services.

For security and maintenance, **Microsoft Defender** will provide antivirus protection, while **AnyDesk** can help the IT staff provide remote support. **7-Zip** will be useful for handling compressed files.

Overall, these software programs cover the main needs of the company's employees and IT department.

Part 4 — Enterprise Network Inventory

Networking Equipment	Quantity	Department/Location	Purpose
ISP Modem	1	Server Room	Provides the internet connection from the ISP
Router	1	Server Room	Manages the connection between the internet and company network

Networking Equipment	Quantity	Department/Location	Purpose
Firewall	1	Server Room	Helps protect the network from unauthorized access
Managed Switch	1	Server Room	Connects wired computers and network devices
Wireless Access Point	2	Office Area	Provides wireless network access
Patch Panel	1	Server Room	Organizes and connects network cables
CAT6 Ethernet Cable	300 meters	Whole Office	Connects computers and network devices
RJ45 Connectors	50 pieces	IT	Used for making Ethernet cable connections

This network equipment is enough for the initial setup of Tyr Technologies' 20-employee office. The **ISP modem and router** will provide internet access, while the **firewall** will help protect the company's network.

The **managed switch** will connect the company's wired devices, while the two access points will provide Wi-Fi coverage around the office. A **patch panel** will keep the network cables organized and easier to manage.

I chose **CAT6 cables** because they are suitable for a modern office network and provide enough capacity for the company's current needs. Extra RJ45 connectors are included so the IT staff can create or replace network cables when needed.

Part 6 — System Administration Roles

Helpdesk Technician

Responsibilities:

- Helps employees with computer and software problems.
- Installs and configures basic software.
- Troubleshoots common hardware and network issues.
- Escalates more difficult problems to other IT staff.

Skills:

- Troubleshooting
- Communication
- Basic networking
- Hardware and software knowledge

Common Tools:

- AnyDesk
- Windows tools
- Ticketing systems
- Remote support tools

Certifications:

- CompTIA A+
- Google IT Support Certificate

Network Administrator**Responsibilities:**

- Manages the company's network.
- Configure routers, switches, and access points.
- Monitors network performance.
- Helps maintain network security.

Skills:

- Networking
- IP addressing
- Network troubleshooting
- Network security

Common Tools:

- Cisco Packet Tracer
- Wireshark
- Command Prompt
- Router and switch management tools

Certifications:

- CompTIA Network+
- Cisco CCNA

Linux System Administrator

Responsibilities:

- Manages Linux servers.
- Installs and maintains server software.
- Manages users and permissions.
- Monitors server performance and security.

Skills:

- Linux commands
- Server management
- Shell scripting
- Troubleshooting

Common Tools:

- Ubuntu Server
- Bash
- SSH
- VirtualBox

Certifications:

- Linux+
- LPIC-1

Cloud Administrator

Responsibilities:

- Manages the company's cloud services.
- Creates and manages cloud resources.
- Monitors cloud systems and performance.
- Helps manage cloud security and access.

Skills:

- Cloud computing
- Networking
- Security
- System administration

Common Tools:

- AWS
- Microsoft Azure
- Cloud management consoles
- Virtual machines

Certifications:

- AWS Certified Cloud Practitioner
- Microsoft Azure Fundamentals (AZ-900)

How They Work Together

The four IT professionals work together to keep the company's technology running properly. The **Helpdesk Technician** handles common employee problems and sends more difficult issues to the appropriate IT staff. The **Network Administrator** manages the company's network and connectivity, while the **Linux System Administrator** manages the Linux server and its services. The **Cloud Administrator** manages the company's cloud resources and services. By working together, they can solve technical problems faster and keep Tyr Technologies' systems available and secure.

Part 7 — Infrastructure Recommendations

Internet Provider

Tyr Technologies should use a **reliable business internet plan with at least 300 Mbps speed**. This should be enough for 20 employees who need internet access for development, communication, cloud services, and other daily tasks.

Server Specifications

The company should use a server with at least:

- Intel Xeon or similar server CPU
- 32 GB RAM
- 2 TB storage

- RAID storage configuration
- Gigabit Ethernet
- Ubuntu Server

These specifications should be enough for the company's current needs and can be upgraded when the company grows.

Backup Strategy

Important company files should be backed up regularly to the **NAS storage**. An additional copy should also be stored on the external backup drives.

The IT department should schedule backups and regularly check if the backups can be restored properly.

Security Recommendations

The company should use a firewall, strong passwords, user access controls, and regular software updates. Employees should also avoid opening suspicious links or downloading unknown files.

Important company information should only be accessible to employees who need it for their work.

Antivirus

Microsoft Defender should be enabled on all Windows computers. It provides basic protection against viruses, malware, and other common security threats.

Password Policy

Employees should use strong and unique passwords for their accounts. Passwords should not be shared with other employees and should be changed if an account is suspected of being compromised.

Multi-factor authentication should also be used for important accounts whenever available.

Expansion Plan

As Tyr Technologies grows, the company can add more computers, network ports, storage, and access points. The server can also be upgraded with more RAM and storage.

The network should be designed with extra capacity so new employees and equipment can be added without needing to completely rebuild the infrastructure.

Overall Recommendation

The recommended infrastructure is suitable for Tyr Technologies as a small startup. It provides the basic equipment, security, backup, and network resources needed for the company's current 20 employees while leaving room for future expansion.

Part 8 — Personal Reflection

Working on this project helped me understand that system administration is not only about fixing computers or managing networks. It also involves planning the IT infrastructure before the company starts using it. I learned how important it is to consider the number of employees in their departments, the type of work they do, and the equipment and software they need.

One of the things I learned from this project was how to create hardware, software, and network inventories. Before doing this project, I mostly thought about computers and networking separately. Now I understand that these components need to work together to support the whole organization. I also learned how different IT roles, such as helpdesk technicians, network administrators, Linux system administrators, and cloud administrators, have different responsibilities but still need to work together.

The most challenging task for me was designing the network infrastructure. I had to think about how the internet, modem, router, firewall, switch, server, access points, printer, and different departments would connect with each other. Creating the network diagram also helped me understand how a real office network can be organized.

Planning is important before deployment because it prevents unnecessary problems and expenses. If a company buys equipment without knowing what it needs, it could end up spending too much money or purchasing equipment that is not suitable for its operations. Proper planning also makes it easier to maintain the network, improve security, and expand the infrastructure in the future.

This project will help me become a better System Administrator because it gave me experience in thinking about an organization's IT needs before implementing them. It also improved my technical documentation, network planning, and problem-solving skills. I learned that a good System Administrator should not only know how to fix problems but should also be able to plan and organize systems properly.

Overall, this project gave me a better understanding of how IT infrastructure is planned for a real organization. It showed me that good preparation and documentation are important for creating a reliable, secure, and manageable IT environment.